
CHRISTOPHER BILSON^{a,*}

ADRIAN GEPP^b

VOLODYMYR NOVYKOV^a

THE TIME SERIES PROPERTIES OF CRYPTOCURRENCY MOMENTS – AN EMPIRICAL APPLICATION OF THE GENERALISED UTILITY OF MOMENTS THEORY

ABSTRACT

In this paper we document and examine the time-series properties of the distribution of a sample of cryptocurrency returns. To accomplish this, we adopt the novel approach of the generalised utility of moments which provides us with a relatively parsimonious, assumption-free means of quantifying the daily distributional properties of each coin. These moments are found to exhibit considerable similarity in their cross-sectional behaviour, particularly in relation to the third moment, the presence of which we interpret as systematic, being robust to the formation of portfolios. This third-moment feature is also found to be present in time-series and is related to both contemporaneous and future returns. Trading strategies designed to exploit this association are found to be capable of generating significant returns and are robust to a broad range of weighting techniques. We also find that simple machine learning techniques can enhance these strategies, the predictability of which is interestingly driven by short-side dynamics.

JEL Codes: G11, G12, G41

Keywords: Cryptocurrency, Distribution moments, Trading strategies

Acknowledgement: This research was supported by a research grant from the Accounting and Finance Association of Australia and New Zealand. We also thank Professor Tom Smith and Dr Milind Tiwari for their valuable feedback.

^a Bond Business School, Bond University, Gold Coast, Queensland, Australia

^b Bangor Business School, Bangor University, Wales, United Kingdom

*denotes corresponding author

THE TIME SERIES PROPERTIES OF CRYPTOCURRENCY MOMENTS – AN EMPIRICAL APPLICATION OF THE GENERALISED UTILITY OF MOMENTS THEORY

ABSTRACT

In this paper we document and examine the time-series properties of the distribution of a sample of cryptocurrency returns. To accomplish this, we adopt the novel approach of the generalised utility of moments which provides us with a relatively parsimonious, assumption-free means of quantifying the daily distributional properties of each coin. These moments are found to exhibit considerable similarity in their cross-sectional behaviour, particularly in relation to the third moment, the presence of which we interpret as systematic, being robust to the formation of portfolios. This third-moment feature is also found to be present in time-series and is related to both contemporaneous and future returns. Trading strategies designed to exploit this association are found to be capable of generating significant returns and are robust to a broad range of weighting techniques. We also find that simple machine learning techniques can enhance these strategies, the predictability of which is interestingly driven by short-side dynamics.

JEL Codes: G11, G12, G41

Keywords: Cryptocurrency, Distribution moments, Trading strategies

1. INTRODUCTION

Financial trading strategies seek to exploit predictable variation in the time series of asset price movements. Expressed generally, this variation arises from either information embedded in fundamental pricing models, such as in arbitrage or equilibrium models of prices (Campbell (2000)), or in deviations from such models, through so-called market anomalies (Jacobs (2015)). In the case of cryptocurrency, the approaches used for traditional asset classes are limited by both the features of the instruments themselves and the manner in which they are traded. Firstly, cryptocurrency has no underlying cash flow, which makes valuation problematic as this characteristic is common to fundamental models of asset valuation (Gordon (1959), Pinto, Robinson et al. (2018)). Instead models of valuation in this context focus on mechanical aspects of coin creation (such as electricity prices (Hayes (2017))) and price behaviour induced by the nature of market participants (Liu, Tsyvinski et al. (2021)).

The latter aspect, being the nature of market participants, has received particular attention in the literature. Cryptocurrency market participants have been documented as having unique characteristics, notably in terms of age and investment experience (Xi, O'Brien et al. 2020) and in their ability to interpret information arrival (Rognone, Hyde et al. 2020). These features cause these markets to be more susceptible to extreme events, such as bubbles (Kyriazis, Papadamou et al. (2020)) and to exhibit different time-series properties from traditional asset classes, such as currencies (Rognone, Hyde et al. 2020). Various theories have been put forward to explain this behaviour, notably salience theory, which proposes that cryptocurrency traders may overweight certain types of information arrival or market movements which capture their attention and therefore overreact to information arrival events (Chen, Lepori et al. (2022) and Smales (2022)).

In aggregate, these features result in the distribution of cryptocurrency returns differing from the Gaussian values associated with normality (or lognormality) that are often assumed in financial models (Karagiorgis, Ballis et al. 2023). Further, heightened levels of investor interest have resulted in these markets experiencing exponential growth, with periods of abnormally high returns consistent with a bubble (Kyriazis, Papadamou et al. (2020)). These periods are associated with high persistence and thereby predictability in both cryptocurrency returns and volatility, similar to the momentum effects observed for equity markets (Rognone, Hyde et al. (2020), Liu, Tsyvinski et al. (2021), Almeida and Gonçalves (2022) and Jia, Liu et al. (2021) and resulting in correlation between successive returns and the higher moments of the distribution. For example, Almeida and Gonçalves (2022) document a number of papers that find evidence of GARCH effects in cryptocurrency returns, consistent with this persistence in volatility. These aspects of the distribution, in combination with the time-series persistence in cryptocurrency returns and volatility potentially serve as a fruitful avenue to quantify the unobservable behavioural aspects of cryptocurrency market participants. In doing so, we seek to investigate the implied trade-off between returns and higher moments, and thereby gain an improved understanding of the pricing of cryptocurrencies and of the modelling of their time-series behaviour.

Research in the area of distribution moments has a long history and is the basis of traditional modern portfolio theory, beginning with Markowitz and Sharpe and Lintner. Their initial model, focusing on mean and variance, was later extended with the 3-moment CAPM of Kraus and Litzenberger (1976) relaxing the assumption of utility being restricted to the first two moments. This model has enjoyed much success, with Harvey and Siddique (2000) showing that there exists a positive and persistent risk premia on market level co-skewness. Subsequent research has shown

that these moments are also related in a conditional sense and hence may be useful in forecasts of future returns (Tay, Christoffersen et al. 2006).

The studies outlined above though are portfolio selection models derived under equilibrium assumptions, where each asset is priced according to its contribution to uncertainty in the investor's end of period wealth, whether that be standard deviation or skewness (the latter being in the case of Kraus and Litzenberger (1976)). These equilibrium assumptions are unlikely to hold in a cryptocurrency setting (not least if these securities are viewed as speculative rather than a store of wealth (as argued by Baek and Elbeck (2015) among others), and as such the price setting process in cryptocurrency markets will result in a different relational form from that derived in these equilibrium-based models (Khuntia and Pattanayak 2018). Further, even should these assumptions be satisfied, quantifying the functional form between expected returns and higher moments requires specification of a specific utility function, which becomes even more complicated once moments above the 3rd are considered.¹ Specifically, under modest HARA assumptions of utility (such as quadratic utility), the assumptions of aversion to risk (standard deviation or variance) and preference for skewness, while subject to debate, are consistent with observed behaviour. The preference for kurtosis though is less obvious from a utility perspective. As such, while papers such as Jia, Liu et al. (2021)) have done much to advance our understanding of these pricing relationships in the cryptocurrency context, there is no certainty that these studies are capturing the true ex ante risk premium. Further empirical issues, such as the spurious association between the moments (MacGillivray and Balanda (1988)), as well as the absence of a functional form for utility make this a challenging area in which to research.

Our approach then is to build upon these initial studies by adopting a more parsimonious distributional and utility framework with assumptions that are more consistent with the functioning of cryptocurrency markets. This provides us with a theoretical yet relatively bias-free alternative to previous approaches in which to investigate the time-series relationship between the higher moments and return. Specifically, our work can be differentiated from previous studies in the area in the following ways. First, we employ intra-day data to compute cross-sectional moment estimates and then use these in time series to investigate their relationship with daily returns. This avoids the spurious association that exists between the moments (as identified by MacGillivray and Balanda (1988)). Second, we use L-moments rather than traditional moments as these have been shown to be more robust in finite sample and less susceptible to outliers making them more suitable for the cryptocurrency context. Finally, we adopt the theoretical approach of the generalised utility of n-moments theory (Billio, Maillet et al. (2022)) to consider the moments independently and in combination. We posit that this framework is not only theoretically sound, but avoids the requirement that the data is normal, that investors hold a diversified portfolio and that cryptocurrency is a reasonable approximation for investor wealth. As such, we believe that our methodology provides a more realistic framework in which to investigate the relationship between returns and higher moments.

¹ Kraus and Litzenberger (1976) ignore moments higher than the 3rd order as the requirement of separability of the utility components is that these higher moment relations with utility be independent.

2. METHODOLOGY AND DATA

Our methodology begins with the traditional financial utility assumption that investor utility derives from the empirical moments of return. Instead of assuming an exact functional form for the investor's utility function, we adopt as our starting point the generalised utility of n-moments approach of Billio, Maillet et al. (2022) which defines (expected) utility as follows;²

$$E[U_i(r_p)] = \sum_{n=1}^N \lambda_{n,i,p} \times m_{n,p}(r_p) \quad \dots(1)$$

$$\text{where } \lambda_{n,i,p} = (-1)^{n+1} n!^{-1} \omega_{n,i} \times g_{n,i} [m_{1,p}(r_p)]^{\tau_N} \quad \dots(2)$$

and where $n = [1, \dots, N]$, $n!$ is the n-factorial and τ_N is a constant.

To illustrate, for the 2nd moment (i.e. variance), the $\lambda_{n,i,p}$ term becomes $\lambda_2 = \left(-\frac{1}{2}\right) \omega_{2,i} \times (g_{2,i} \times m_1)$.³ The first term reflects that the premium on variance is negative (reflecting risk aversion), with the second term being a weighting function capturing the impact of variance on utility and the final term being how this moment (variance) is linked to the first moment (return), akin to a risk premium. Assuming initially a constant weighting structure across individuals and time, equation (1) can be aggregated across the first four moments to produce the Generalised Utility of N-Moments (GUN) measure.

$$E(U_p) \cong \sum_{j=1}^4 \lambda_j \times m_j \quad \dots(3)$$

where λ_j is a column vector of weights for each of the four moments and m_j is the empirical value of the j-th moment.

This specification then represents utility as an aggregate of the empirical properties of “greediness”, via the first moment; “risk aversion” via the second moment; and “prudence” and “temperance” for the third and fourth moments, respectively. The specification in equation (3) therefore, uniquely allows us to incorporate the previously mentioned stylised facts relating to cryptocurrencies, providing a measure of utility (which could be used to rank portfolios), but which also *jointly* capture the distribution of returns. As mentioned above, for robustness reasons, we adopt the L-moment approach of Bernard, El Mekkaoui De Freitas et al. (2019) when calculating the empirical moments, which has been shown to be more robust in finite samples, viz;

$$\left\{ \begin{array}{l} m_1 = E(r_{[1:1]}) \\ m_2 = (2!)^{-1} E(r_{[2:2]} - r_{[1:2]}) \\ m_3 = (3!)^{-1} E(r_{[3:3]} - 2r_{[2:3]} + r_{[1:3]}) \\ m_4 = (4!)^{-1} E(r_{[4:4]} - 3r_{[3:4]} + 3r_{[2:4]} - r_{[1:4]}) \end{array} \right\} \quad \dots(4)$$

² Note that Billio et al (2022) include an additional term containing a remainder (due to their use of the Lagrangian/factorial in their derivation), and which we exclude here as it doesn't pertain to nor impact the present paper.

³ Consistent with Scott and Horvath (1980), the relationship between utility and the moments will be positive (negative) for odd (even) moments.

where $m_1(.)$, $m_2(.)$, $m_3(.)$ and $m_4(.)$ are the L-mean, L-variance, L-skewness and L-kurtosis measures, respectively.⁴

3. SAMPLE AND DESCRIPTIVE STATISTICS

To compute our empirical moments, we obtain hourly price data on 50 cryptocurrencies over the 18th of August, 2017 to the 21st of July, 2021 period, sourced from Binance.⁵ The cryptocurrencies chosen represent the largest and most liquid traded over this period, the sample for which was selected so as to make the results of the paper relevant to cryptocurrency traders generally. For each coin, on each day, the hourly returns are used in cross section to compute daily values of the m_1 , m_2 , m_3 and m_4 for each coin. Our method is therefore unique in this regard, in that we use intra-day data to generate daily values for each moment for each coin. This avoids the typical approach, where daily data is used in time-series to compute the moments which creates a spurious association between the moments from the use of overlapping observations in generating the time-series of moments in addition to the fact that the moments themselves are not independent (MacGillivray and Balanda 1988).

Descriptive statistics for the currencies in our sample are reported in Table 1. Column (1) of Table 1 reports the ticker symbol for each cryptocurrency, with the averages of the daily values for each of the L-moments (m_1 , m_2 , m_3 and m_4) in columns (2) through (5). Our sample period is characterised by a bullish sentiment, with only two of the 50 coins exhibiting a negative mean return. Both the L-moment for variance (m_2) and kurtosis (m_4) exhibit significant variation between coins, with the lowest variance being 0.0042 and the highest 0.0143, with a similar large range for kurtosis, being from 0.0010 to 0.0036, suggesting significant coin-specific behaviour. However, we find strong evidence of a systematic pattern in the skewness behaviour with over 90% of the coins exhibiting a positive third L-moment (m_3). In summary, the L-moments indicate that the return series are categorically non-normal, and Jarque-Bera tests of this null hypothesis (reported in column (6)) are rejected at conventional levels of significance for all coins. In terms of time-series behaviour, all series exhibit negative first order serial correlation (which is prima-facie evidence of mean-reversion at the daily frequency). None of the returns appears to be non-stationary though and ADF tests (column (8)) reject the unit root null hypothesis for all coins in the sample, consistent with difference stationarity in the underlying price series.

⁴ Some interesting, related works, find evidence that the ratio of moments (specifically derivatives of the utility function) might also be important (similar to intuition underlying the Sharpe-ratio, as discussed in Lo (2002). Crainich and Eeckhoudt (2008) for example, introduce the concept of equivalency prudence, being a risk aversion from an arbitrage position between values of the first and third moments. Lajeri-Chaherli (2004) extend the four moment framework considered here to include “edginess”, being the fifth-order, whilst Caballé and Pomansky (1996) expand this to a more general N -th order framework.

⁵ <https://www.binance.com/en-AU>.

Table 1 - Descriptive statistics for individual cryptocurrencies

(1) <i>Symbol</i>	(2) <i>m1</i>	(3) <i>m2</i>	(4) <i>m3</i>	(5) <i>m4</i>	(6) <i>J-B</i>	(7) <i>p(1)</i>	(8) <i>ADF</i>
ADA	0.0001	0.0061	0.0000	0.0012	260095.17	-0.0518	-32.16***
ALGO	0.0000	0.0082	0.0001	0.0015	166257.75	-0.0621	-28.71***
ATOM	0.0001	0.0073	0.0001	0.0013	348906.22	-0.0480	-28.47***
AVAX	0.0024	0.0121	0.0021	0.0036	8826209136.33	-0.0580	-17.86***
BCH	0.0001	0.0058	0.0000	0.0013	610349.16	-0.0303	-25.32***
BNB	0.0003	0.0061	0.0000	0.0012	1366907.13	-0.0498	-32.08***
BTC	0.0001	0.0042	0.0000	0.0010	1172024.42	-0.0551	-34.36***
BTG	-0.0004	0.0111	0.0002	0.0026	242434.81	0.0319	-13.56***
CELO	0.0011	0.0119	0.0014	0.0030	96583466.20	-0.1583	-16.65***
DASH	0.0001	0.0058	0.0001	0.0012	4619417.18	-0.0293	-27.64***
DCR	0.0003	0.0069	-0.0001	0.0014	2079153.36	-0.0131	-21.31***
DGB	0.0003	0.0093	0.0000	0.0017	56845.59	-0.0496	-20.37***
DOGE	0.0003	0.0063	0.0002	0.0014	26793261.53	-0.0142	-23.02***
DOT	0.0004	0.0085	0.0002	0.0016	2105814.35	-0.0442	-20.57***
EGLD	0.0004	0.0089	0.0001	0.0017	61160.71	-0.0372	-19.28***
EOS	0.0000	0.0055	0.0000	0.0013	952255.80	-0.0354	-31.72***
ETC	0.0001	0.0057	0.0000	0.0013	1633854.33	-0.0571	-30.18***
ETH	0.0001	0.0053	0.0000	0.0012	1484730027.78	-0.0513	-33.59***
FIL	0.0011	0.0084	0.0012	0.0026	9120222606.15	-0.5583	-17.28***
HBAR	0.0002	0.0081	0.0002	0.0017	12509073.95	-0.0308	-23.53***
HNT	0.0008	0.0108	0.0007	0.0023	13468963.05	-0.0624	-20.10***
ICP	-0.0003	0.0143	0.0008	0.0026	293113.62	0.0378	-12.31***
KLAY	0.0012	0.0073	0.0001	0.0014	15145.57	-0.0188	-9.53***
KSM	0.0003	0.0103	0.0002	0.0018	58610.10	-0.0418	-19.54***
LTC	0.0000	0.0058	0.0001	0.0013	645266.05	-0.0539	-33.56***
LUNA	0.0003	0.0086	0.0003	0.0020	162847.88	0.0341	-20.31***
MATIC	0.0005	0.0091	0.0004	0.0019	468394294.41	-0.0430	-27.45***
MDX	0.0004	0.0075	0.0007	0.0023	98651.71	-0.0639	-11.81***
NEAR	0.0002	0.0102	0.0002	0.0019	26449.26	-0.0438	-19.34***
NEO	0.0001	0.0068	0.0001	0.0013	795876.99	-0.0488	-34.29***
ONE	0.0002	0.0090	0.0002	0.0017	488426.27	-0.0616	-27.35***
ONT	0.0000	0.0069	0.0001	0.0014	634426.19	-0.0426	-31.53***
QTUM	0.0000	0.0065	0.0000	0.0014	1123759.35	-0.0361	-32.80***
RVN	0.0001	0.0070	0.0001	0.0015	468145.22	-0.0191	-24.98***
SC	0.0003	0.0087	-0.0001	0.0017	115124.78	-0.0290	-19.81***
SOL	0.0005	0.0102	0.0002	0.0019	55885.44	-0.0248	-19.98***
STX	0.0002	0.0088	0.0002	0.0019	4073240.27	-0.0109	-24.80***
TFUEL	0.0002	0.0086	0.0002	0.0020	1938312.38	-0.0476	-25.97***
THETA	0.0003	0.0080	0.0000	0.0014	166893.33	-0.0516	-29.14***
TRX	0.0001	0.0057	0.0000	0.0012	1152168.55	-0.0403	-31.15***
VET	0.0001	0.0072	0.0001	0.0014	3374981.10	-0.0567	-31.68***
WAVES	0.0001	0.0068	0.0002	0.0014	744248.68	-0.0310	-27.60***
XEM	0.0001	0.0097	-0.0002	0.0019	69119.55	-0.0927	-17.84***
XLM	0.0001	0.0059	0.0001	0.0012	1365361.57	-0.0513	-31.26***
XMR	0.0001	0.0053	-0.0001	0.0010	1793161.74	-0.0310	-29.57***
XRP	0.0002	0.0055	0.0002	0.0013	334969970.04	-0.0503	-29.43***
XTZ	0.0001	0.0072	0.0000	0.0014	1566610.18	-0.1023	-27.68***
ZEC	0.0001	0.0062	0.0001	0.0012	1911911.31	-0.0704	-28.68***
ZEN	0.0003	0.0093	0.0003	0.0018	238657.28	-0.0255	-20.61***
ZIL	0.0001	0.0075	0.0000	0.0014	183799.11	-0.0474	-29.05***

Note: This table presents descriptive statistics for each of the 50 cryptocurrencies in our sample. The values in the table represent daily figures and statistics, where each value is calculated from intra-day data for each cryptocurrency. Columns (2) through (5) report the L-moments (as per equation 4), where m1, m2, m3 and m4 denote the first (mean); second (variance); third (skewness) and fourth (kurtosis) L-moment. Column (6) displays the Jarque-Bera statistic for normality (which is distributed as a chi-squared with N-2 degrees of freedom under the null hypothesis of normality), and columns (7) and (8) the first order auto-correlation term (p_1) and Augmented Dickey-Fuller tests on the cryptocurrency returns, respectively. ***/*** denotes statistical significance at the 10%/5%/1% level.

Following on from the documentation of the individual coin behaviour in Table 1, we aggregate across the 50 cryptocurrencies to form portfolios in order to evaluate the degree of commonality in the behaviour of the L-moments over time. This focus on portfolios will be a common theme in this paper as it is the common or systematic behaviour which we are seeking to document and analyse. From the perspective of finance theory, any coin-wise features that are robust to the formation of portfolios are common to coins generally and there are considered non-diversifiable and therefore systematic.

These results are presented in Table 2, the Panel A of which reports descriptive statistics on the time series behaviour of each of the four moments in aggregate, where each moment is calculated as a daily value from the intra-day data on each day, and then aggregated using either an equal, size or volume weighting. All three weighting methods indicate positive values across all four moments, consistent in principle with a positive risk-return relationship, as well as with a distribution that contains both a positive skew and is leptokurtic.⁶ We observe a similar rejection of normality as with the coin-wise statistics in Table 1, and with the observance of negative serial correlation.

In conclusion, three features emerge at this point. First, that the observable patterns in the return distributions appear to be generally robust to the coin-weighting method (ie equal, volume or size).^{7,8} Second, aggregation by portfolio fails to remove the distributional features observed at the coin level, indicating that these patterns are, at least to some degree, common to all coins and thereby systematic (non-diversifiable). Third, given the consistency of the moments to how they are weighted by coin, we adopt an equal-weighting approach by coin for the remainder of the paper.⁹

Table 2: Descriptive statistics for cryptocurrency moments in aggregate

(1) <i>Weighting method</i>	(2) <i>m1</i>	(3) <i>m2</i>	(4) <i>m3</i>	(5) <i>m4</i>	(6) <i>J-B</i>	(7) <i>p(1)</i>	(8) <i>ADF</i>
<i>Equal</i>	0.0002	0.0068	0.0001	0.0015	10,284,829.88***	-0.0399***	-35.44***
<i>Size</i>	0.0004	0.0059	0.0001	0.0013	225,911.62***	-0.0484***	-34.97***
<i>Volume</i>	0.0001	0.0042	0.0000	0.0010	250,145.80***	-0.0462***	-34.88***

Note: This table presents descriptive statistics for three weighted portfolios (equal, size and volume) aggregated across the 50 cryptocurrencies in our sample. The values in the table represent daily figures and statistics, where each value is calculated from intra-day data for each cryptocurrency. Columns (2) through (5) report the L-moments (as per equation 4), where m1, m2, m3 and m4 denote the first (mean); second (variance); third (skewness) and fourth (kurtosis) L-moment. Column (6) displays the Jarque-Bera statistic for normality (which is distributed as a chi-squared with N-2 degrees of freedom under the null hypothesis of normality), and columns (7) and (8) are the first order auto-correlation term (p_1) and Augmented Dickey-Fuller t-test on the cryptocurrency returns, respectively. */**/** denotes statistical significance at the 10%/5%/1% level.

⁶ It should also be highlighted that while the skewness value here is quite low in magnitude, these are daily returns and are also L-moments, which are smaller in magnitude than traditional moments (see discussion in Billio et al, 2022).

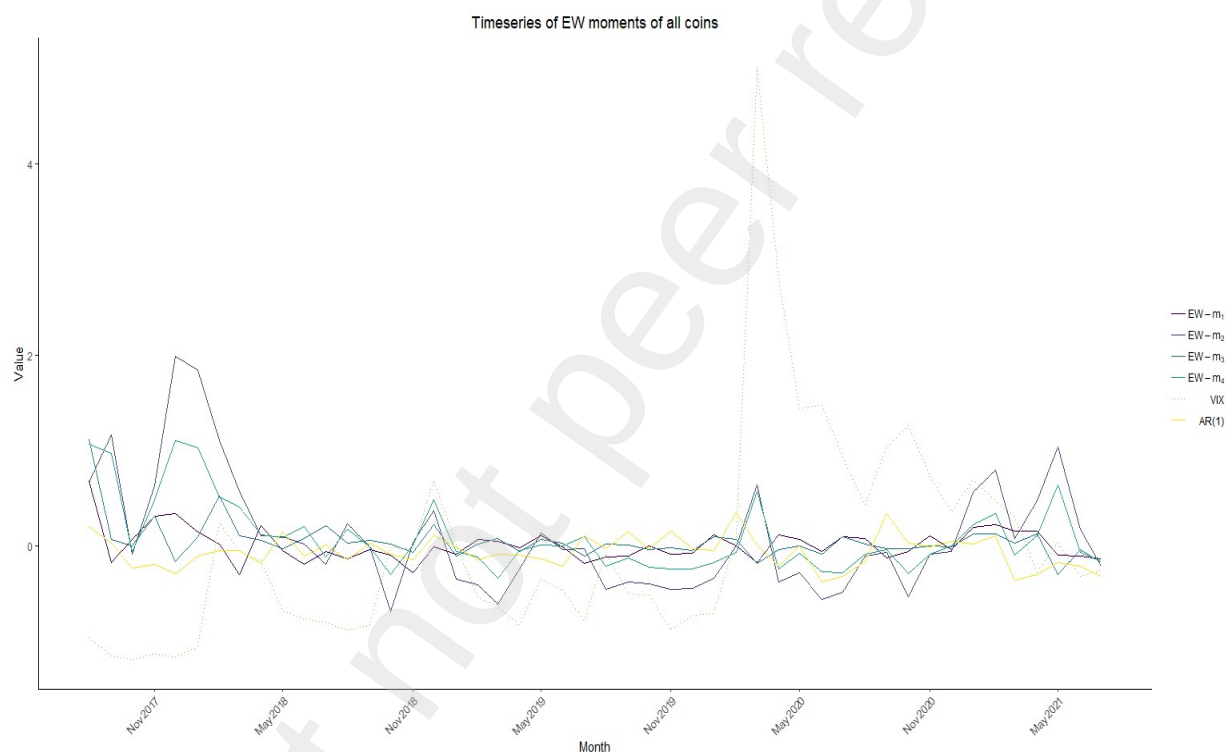
⁷ While we revisit the issue of portfolio formation more formally in the results section of the paper, we feel that it is important to establish that the trading volumes and sizes of the coins do not appear to affect the generalisability of the results, with the cryptocurrencies all displaying generally consistent behaviour in cross-section.

⁸ We leave the possibility that the results of the paper are further nuanced by issues of trading volume and size to future research, and we avoid investigation of this issue at this point to limit any perceptions of data mining.

⁹ The results (available from the author(s) on request) indicate that the size and volume-weighted results are consistent with that of the equal-weighted portfolio reported in the table.

Figure 1 reports the time series behaviour of each of the four equal-weighted moments from Table 2. The time-series behaviour is telling as it shows the flow of information and investor trading behaviour on the distributional properties of cryptocurrency returns and we believe we are the first to document this behaviour in this way. Specifically, we find that while there are some periods where the moments are clustered and positively related (see November-December of 2018 for example), the skewness (in particular) often moves in opposition to the other moments (November-December 2017 and May 2021).¹⁰ As a final, yet preliminary investigatory tool, we also plot the rolling monthly AR(1) coefficient of the return, as a proxy for momentum/predictability as well as the VIX. We find no obvious pattern between either the VIX of the AR(1) coefficient for any of the four moments, indicating that the time-series behaviour we are investigating is neither due to equity market spillovers (whether that be due to portfolio rebalancing or wealth effects) nor attributable to daily momentum in cryptocurrency returns.¹¹

Figure 1: Time series behaviour of moments, equal-weighted by coin



Note: This figure presents the time series of the four moments, equally weighted by coin, across each of the days in the sample in addition to the VIX and a rolling serial correlation on the EW cryptocurrency portfolio. The values in the table are computed using daily L-moment figures, where each daily moment is calculated from intra-day data for each cryptocurrency and normalised to allow for improved comparability.

¹⁰ At this point we also highlight that our method is unique in that we use intra-day data to generate daily values for each moment for each coin. This avoids the typical approach, where daily data is used in time-series to compute the moments such that there is a spurious association that is created between the moments from the use of overlapping observations in generating the time-series of moments. Secondly, as explained, any pattern in the time-series of moments will be either due to the flow of information arrival and its properties or the subsequent behaviour of traders. While beyond the scope of the present paper, a fruitful line of inquiry would be in ascertaining the source and nature of information events that lead to asymmetric distribution moments influences.

¹¹ Specifically, the VIX and moment portfolios are more or less independent, with the correlations between the VIX and the EW-m1, EW-m2, EW-m3 and EW-m4 moments being -0.021, -0.034, -0.048 and -0.001, respectively.

In Table 3, we now expand on the figure 1 observations to consider the time-series properties of the moments in more detail. Panel A of Table 3 extends the Table 2 descriptive statistics by considering the time-series and correlational structure for each of the four L-moments. To ensure comparability, the same equal weighting by coin to that used in Table 2 is adopted. Interestingly, the 1st and 3rd moments all exhibit fluctuations of similar magnitude, with the 2nd and 4th moments being (approximately) larger by around 40 and 10 times, respectively than the 1st. Further, the 1st and 3rd moments display almost no persistence but the 2nd ($\rho = 0.6637$, p-value = 0.00) and 4th ($\rho = 0.1993$, p-value = 0.00) moments display significant positive time-series predictability. This latter result is consistent with the observation of GARCH effects in studies such as Sapuric, Kokkinaki et al. (2020) and as summarised in the literature review of Almeida and Gonçalves (2022).

Table 3: Panel A – Descriptive statistics for equal-weighted moment portfolios

(1) <i>L-moment</i>	(2) <i>Mean</i>	(3) <i>Variance</i>	(4) <i>p(1)</i>
<i>m1</i>	0.000156	0.000005	-0.0344
<i>m2</i>	0.006817	0.000015	0.6637***
<i>m3</i>	0.000133	0.000001	0.0135
<i>m4</i>	0.001468	0.000002	0.1993***

Note: This table presents descriptive statistics for each of the four L-moments, computed by equally-weighting the moments across the 50 cryptocurrencies in our sample. The moments are computed using L-moments, where each daily moment is calculated from intra-day data for each cryptocurrency.

Panel B of Table 3 reports correlations in the time-series of the moments. To control for the impact of influential observations (days) we use both parametric (Pearson) and non-parametric (Spearman rank) correlations. The above (below) diagonal represents the Pearson (Spearman rank) correlations. Consistent with Panel A, we also observe similarity between the time-series behaviour of the 1st and 3rd moments (0.60 and 0.58) and between the 2nd and 4th moments (0.78 and 0.84). Of particular interest is that there is a significant magnitude effect driving a relationship between the 3rd and 4th moments, which is strongly positive using parametric (0.38) but which disappears in the non-parametric results, when the effect of influential observations is removed (-0.01). Further, in relation to their association with return (the 1st moment), skewness and kurtosis are both positively related (0.60 and 0.13, respectively). However there is little evidence of a cross-sectional risk-return relationship (between *m1* and *m2*) for the cryptocurrencies in our sample (which is 0.03 and 0.02 for the parametric and non-parametric metrics, respectively). This finding adds to the findings of Smales (2022) on the different relationship that exists between cryptocurrency returns and measures of uncertainty, compared with traditional asset classes.

Table 3: Panel B – Cross correlations of equal-weighted moment portfolios

	<i>m1</i>	<i>m2</i>	<i>m3</i>	<i>m4</i>
<i>m1</i>	1.00	0.02	0.58	0.01
<i>m2</i>	0.03	1.00	0.00	0.84
<i>m3</i>	0.60	0.16	1.00	-0.01
<i>m4</i>	0.13	0.78	0.38	1.00

Note: This table presents descriptive statistics for three weighted portfolios (equal, size and volume) aggregated across the 50 cryptocurrencies in our sample. The values in the table are computed using daily L-moment figures, where each daily moment is calculated from intra-day data for each cryptocurrency. Pearson (parametric) correlations are reported in the lower diagonal, while spearman (non-parametric) correlations are reported above the diagonal.

Finally, we consider the moments in combination and examine the association of these combinations with the first moment (the mean return). Consistent with the previous analysis, we adopt an equal-weighting approach when combining the moments, which provides us with five combinations; $m_{2,3}$, $m_{2,4}$, $m_{3,4}$ and $m_{2,3,4}$. Consistent with Panel B, the association between m_3 and m_1 dominates the results, with the highest correlations all involving m_3 . In contrast, the combination measuring symmetric dispersion ($m_{2,4}$) is almost uncorrelated with the mean return (m_1). As a final point, we are reasonably confident that these findings are robust to the effect of influential observations or outliers, as these findings persist whether we measure using parametric or non-parametric correlations.

Table 3: Panel C – Cross-correlations of equal-weighted moment combinations

(1) <i>Equal-Weighted combination</i>	(2) <i>Parametric (Pearson)</i>	(3) <i>Non-parametric (Spearman)</i>
$m_{2,3}$	0.32	0.36
$m_{2,4}$	0.02	-0.01
$m_{3,4}$	0.36	0.43
$m_{2,3,4}$	0.23	0.23

Note: This table presents contemporaneous correlations of the first L-moment (return) with portfolios of moment combinations for each of for four different weighting methods. Note that the daily L-moments for each cryptocurrency is calculated from intraday data and is equally weighted by coin. Column (2) displays the parametric correlation (Pearson) and column (3) the non-parametric (Spearman).

In summary, the Table 3 results (Panels A, B and C) are consistent with the findings of Rognone, Hyde et al. (2020) and demonstrate that traditional finance assumptions regarding parametric risk relationships (i.e. between the 1st and 2nd moments) are less reliable in a cryptocurrency context than what we observe in traditional asset classes. These results do suggest though that skewness is an important feature of cryptocurrency returns and that this finding is consistent with the theoretical propositions that we outlined at the front of the paper. Having documented the empirical behaviour of the moments and their statistical association both individually and in combination with contemporaneous returns, we now investigate the economic content of the moments and consider whether profitable trading strategies from this information are feasible.

4. RESULTS

The central thesis of our analysis then is whether there is information content in the moments of the distribution, as the theory outlined earlier (in particular Kraus and Litzenberger (1976) and Billio, Maillet et al. (2022)) purports that it is the distribution (represented by the moments jointly) which best approximates investor utility. Following on from this discussion, the most

direct way of establishing this empirically then is to determine whether this information flows through to subsequent price changes (returns).

We adopt as our framework for providing a theoretical explanation of this relationship the seminal paper of Hong and Stein (1999) who theorise that markets are fragmented into market participants who act instantaneously to news arrival which is impounded immediately into asset returns. Other investors, however, who do not have access to this information will instead follow a return-chasing strategy (such as the well-documented momentum effect of De Bondt and Thaler (1985) and subsequently formalised as a trading strategy by Jegadeesh and Titman (1993)) and by their behaviour impart a pattern in the moments in time-series. We theorise therefore that this second class of investors do not simply use the information in past returns (as prior momentum studies argue)¹², but rather use information from the past patterns in the whole distribution (i.e. of all four moments) of past returns. That is, as these moments are related to utility, investors will allocate funds when the utility is favourable given their preferences over these four moments. We hypothesise therefore that this suggests a relationship between past moments and future returns exists, which is measurable to the extent that a conditional trading strategy is able to generate returns above some naive benchmark.

The intuition then underlying the trading strategy is then to exploit an inherent return predictability be within the time-series movement in the moments, and to thereby adjust the traditional momentum strategy from predicting $m_{1,t+1}$ using just $m_{1,t}$ to now expanding the conditioning information to be $f(m_1, m_2, m_3, m_4)$ where the latter term represents the preferences (utility) of cryptocurrency market participants. In our research it is not possible for us to disentangle the information and momentum effects of Hong and Stein (1999), and instead seek to identify whether the aggregate behaviour of these investor groups is consistent with our known understanding of utility theory in relation to empirical moments of cryptocurrency returns.

To obtain our utility measure, for each coin on each day, we initially adopt an equally weighted average of the four moments (m_1 , m_2 , m_3 and m_4). For example, when all four moments are used, the following weighting structure applies;¹³

$$U = E(U) \cong \sum_{j=1}^4 \lambda_j \times m_j \quad \dots(5)$$

$$\text{where } \lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 0.25$$

This produces eleven equal-weighted utility series (i) $m_{1,2}$; (ii) $m_{1,3}$; (iii) $m_{1,4}$; (iv) $m_{2,3}$; (v) $m_{2,4}$; (vi) $m_{3,4}$; (vii) $m_{1,2,3}$; (viii) $m_{1,2,4}$; (ix) $m_{1,3,4}$; (x) $m_{2,3,4}$ and (xi) $m_{1,2,3,4}$. As such, on days when data is available for all 50 coins, we will have a daily utility value for each coin on each of the 1,399 days in the sample. Note also, as observed from Table 2, that due to differences in magnitudes, we normalise each of the moments by subtracting the mean and dividing through by the standard

¹² To be clear, while the traditional momentum studies focus on returns, more recently risk (variance) has been employed as a means of either enhancing this strategy or explaining the magnitude of the momentum strategies success. Similarly, studies on Garch-in-mean effects (see discussion in Bollerslev et al, 1992) model the relation between the serially correlated second moment (variance) on the expected value (mean) of returns. The present study can then be thought of as what we see as a natural extension of the intuition underlying these approaches beyond the second moment.

¹³ Note that Billio et al (2022) include an additional term containing a remainder (due to their use of the Lagrangian/factorial in their derivation), and which we exclude here as it doesn't pertain to nor impact the present paper.

deviation. This equal-weighted approach serves as a naïve benchmark for evaluating a range of alternative weighting functions.

The second weighting method follows the Generalised Utility of N-Moments (GUN) weighting approach. That is, for each coin, for each day, we re-apply equation (5) with the weights as per espoused by Billio, Maillet et al. (2022), which are $\lambda_1 = 1$, $\lambda_2 = 0.5$; $\lambda_3 = 0.25$; $\lambda_4 = -0.1$. As with the equal-weighted method, we consider all combinations of the moments, resulting in eleven GUN weighted utility series.

Our 3rd weighting method follows that of Fama and French (1992) which uses a terzile sorting method whereby on each day we sort each moment according to 3 bins (top 30%, middle 40% and bottom 30%). The value for that day, for that particular moment, is the difference between the mean values in the top and bottom bins. For example, on day t , the m_3 measure is the mean of the m_3 measure in the top 30% of m_3 values less the mean of the m_3 measure in the bottom 30% of m_3 values. Again, we consider all combinations of the moments, resulting in four Fama and French (FF) weighted utility series arising from a double sort on (i) $m_{1,2}$; (ii) $m_{1,3}$; (iii) $m_{1,4}$; (iv) $m_{2,3}$; (v) $m_{2,4}$; (vi) $m_{3,4}$; a triple sort on (vii) $m_{1,2,3}$; (viii) $m_{1,2,4}$; (ix) $m_{1,3,4}$ and (x) $m_{2,3,4}$ and a quadruple sort on (xi) $m_{1,2,3,4}$.

Our final approach uses machine learning to determine the weighting function. In doing so, we are careful to ensure our motivations are consistent with the focus of the paper, being to use the most parsimonious, bias-free approach available.¹⁴ More specifically while we seek to employ machine learning techniques to potentially arrive at a better approximation of utility (through a more optimal weighting of the moments than that formed by using equal, GUN or Fama and French-based weights), it is important that we ensure that the ability of any of the weighting approaches to predict future returns be due to inherent time-series association within the moments and future returns, rather than confounding this by embedding a predictive component within the methodology.¹⁵

Our approach then is to apply a linear model, akin to linear regression (with a suppressed intercept), where we provide as inputs values for each of the four moments for the current day and the daily return for the equal-weighted index of coins for the following day. We employ a linear activation function, which is deliberately chosen to be in the same form as for the other weighting schemes outlined above;

$$EW_{t+1} = \text{activation function} \left(\sum_{n=1}^4 w_n \times m_{n,t} \right) \quad \dots(6)$$

This approach produces a reasonably stable set of weights (w_n) and introducing additional complexity (via additional input layers and varying the activation function) fails to produce a significant difference in the model parameters. Similarly, this choice has little impact on the

¹⁴ For example, we leave the issue of time-variation to further research and keep these weights static to be consistent with the other allocation methods (that is the data (moments) will change over time, but the way in which they are weighted are unconditional with respect to time).

¹⁵ Our approaches are therefore deliberately naïve and unconditional, and while we see conditional (rebalanced) strategies employing more complex predictive functions as a fruitful line of empirical inquiry, this is beyond the scope of the present paper, and we leave this for future research.

model performance, as represented by the MSE loss values, which we attribute to the simplicity of the model specification in (6) above.¹⁶

While technically the ML procedure produces daily weights, we constrain these to be unconditional to ensure consistency with the other four weighting techniques (which are also time-invariant and therefore unconditional). To obtain our final model, we average out the weights obtained through a set of ten cross-validated models to arrive at the final set of weights per moment.¹⁷ After normalising we obtain $\lambda_1 = 0.14$, $\lambda_2 = -0.40$; $\lambda_3 = 0.57$; $\lambda_4 = 0.30$, which are notably consistent with the time-series behaviours we observed earlier. Due to the negative serial correlation in returns, there is a much lower weight on the first moment and a much higher weight on the 3rd due to its strong relationship with subsequent returns. Interestingly, the theoretical weight on variance (-0.50) which is a common feature of many financial models is very similar to the coefficient of -0.40 obtained through the machine learning approach. The coefficient on the 2nd moment (variance) is consistent with a growing body of literature that finds that cryptocurrencies with higher variance display lower returns in future (Lee and Wang 2024).

We therefore have five weighting functions from which to evaluate the information present in the distribution moments; the individual moments which we term the pure moment weights (PM), equal-weights (EW); Generalised Utility of Moment weights (GW); Fama and French weights (FF) and Machine learning weights (ML). In investigating our trading strategy, we initially consider all coins in aggregate (i.e. form portfolios across all coins) and then apply the same strategies coin-wise. In our analysis, we employ a conditional trading strategy that only uses information available to investors at the close of the previous day. Our initial strategy takes the position of an investor who takes either a long or short position in a portfolio of cryptocurrencies based on a signal they receive on the previous day. If the signal is high, then the investor places a long position in the equal-weighted cryptocurrency index on the next day and if the signal is low a short position is placed. The unconditional mean for the signal over the sample period is used to determine a high or low signal.¹⁸

To illustrate, using the GUN-weighted signal (GW) formed using moments m_1 and m_2 , the investor would calculate a weighted average of the equal-weighted m_1 and m_2 moments for each trading day in the sample using the GUN weights, and also the unconditional mean of this signal over the sample period. For each day that the signal was higher (lower) than the unconditional mean, a long (short) position in the equal-weighted cryptocurrency index is taken on the following day. The portfolio is then rebalanced for each day across the sample and the cumulative return reported in Panel A of Table 4. To evaluate the profitability of these strategies, we use a risk-free rate and the long-only holding period return from the equal-weighted index of the 50 cryptocurrencies as a benchmark. The coin-wise strategy is identical, except that the inputs to the

¹⁶ For example, both RELU and Tanh activation functions were employed to investigate the impact of non-linearity on the estimated weights and model performance. However, we opted for the linear activation function in (6), as this was both consistent with the underlying theory and comparable with the other weighting functions mentioned earlier, with only a marginal improvement in model performance from the incorporation of non-linearity. As a final form of sensitivity, all results were re-estimated using MAE as a loss function with very little impact on the estimates. All sensitivity results are available from the author(s) on request.

¹⁷ We also use the accuracy results (obtained per epoch) to determine the number of epochs with which to train the final model before testing it. We then combine the train, using the final train to validate the data on the whole non-out-of-sample dataset available, and finally test the produced model on the holdout data.

¹⁸ The strategy, being long-short is not technically zero cost, but could be amended to be so. In this paper, we have not followed this approach as our objective was to keep any bias from the trading strategy itself to be as minimal as possible.

strategy are done using as inputs the four L-moments (m_1 , m_2 , m_3 and m_4) *for each coin* and the allocation is then done on the *returns of that coin* on the following day, rather than being allocated in the equal-weighted index, as with the aggregate strategies.

Table 4 – Panel A: Conditional trading strategies using lagged moment combinations

	(1) Utility Metric (U)	(2) Aggregate Long-short (p.a.)	(3) Coin-wise Long-short (p.a.)
Panel A: PM	m ₁	-0.0075	0.0012
	m ₂	0.0026	0.0099
	m ₃	0.0092	0.0111
	m ₄	-0.0078	0.0038
Panel B: EW	m _{1,2}	0.0043	0.0086
	m _{1,3}	0.0179*	0.0145
	m _{1,4}	0.0019	0.0064
	m _{2,3}	-0.0103	0.0027
	m _{2,4}	-0.0053	0.0058
	m _{3,4}	-0.0072	0.0030
	m _{1,2,3}	0.0041	0.0086
	m _{1,2,4}	-0.0019	0.0063
	m _{1,3,4}	0.0118	0.0113
	m _{2,3,4}	-0.0008	0.0072
	m _{1,2,3,4}	-0.0112	0.0013
	m _{1,2}	0.0084	0.0092
Panel C: GW	m _{1,3}	0.0080	0.0088
	m _{1,4}	-0.0074	0.0012
	m _{2,3}	-0.0081	0.0010
	m _{2,4}	-0.0139	-0.0020
	m _{3,4}	0.0098	0.0116
	m _{1,2,3}	0.0045	0.0074
	m _{1,2,4}	0.0055	0.0077
	m _{1,3,4}	0.0017	0.0057
	m _{2,3,4}	-0.0084	0.0008
	m _{1,2,3,4}	0.0008	0.0055
	m _{1,2}	-0.0130	-0.0101
	m _{1,3}	-0.0348	-0.0141
Panel D: FF	m _{1,4}	-0.0255	-0.0070
	m _{2,3}	-0.0149	-0.0137
	m _{2,4}	-0.0014	0.0062
	m _{3,4}	-0.0428	-0.0208
	m _{1,2,3}	-0.0156	-0.0106
	m _{1,2,4}	-0.0215	-0.0087
	m _{1,3,4}	-0.0153	-0.0060
	m _{2,3,4}	-0.0020	-0.0024
	m _{1,2,3,4}	-0.0277	-0.0113
	m _{1,2,3,4}	0.0185*	0.0148
Panel E: ML			

Note: This table presents contemporaneous correlations of the first L-moment (return) with portfolios of moment combinations for each of four different weighting methods. Note that the daily L-moments for each cryptocurrency is calculated from intraday data and is equally weighted by coin. Panel A reports the Pure moment (PM) portfolios, whereby the weight in each moment is 100%, while Panel B weights each moment equally (EW), as per equation (5). Panel C uses the GUN weights (GW) of Billio et al (2022) (as per equation (6)), Panel D uses the Fama and French (FF) double and triple sorting procedures to form the moment portfolios while Panel E uses the machine learning (ML) approach to determine the weights on each moment. The symbol * indicates that the return for the strategy is in excess of the risk-free rate, while ** indicates that the return is in excess of the buy and hold position on the equal-weighted cryptocurrency portfolio.

The aggregate trading strategy results presented in Panel A of Table 4 are wholly unimpressive. For the aggregate results, only 15 of the 42 trading strategies (36%) produce a positive return with only two strategies (the equal-weighted $m_{1,3}$ combination and the machine learning weights) in excess of the risk-free rate and none that exceed the buy and hold return benchmark. The Coin-wise strategies perform much better, with 28 strategies (or 67%) being profitable, but these returns are all quite small and none of these are in excess of either the risk-free or buy and hold benchmarks.

This aggregated long-short approach from Table A however confounds the long and short effects and a better understanding of these strategies can be obtained from isolating these two influences. Our approach is therefore similar to the signed decomposition approach of Lee and Wang (2024), however we use a trading strategy framework in contrast to their variance decomposition methodology. Panel B of Table 4 therefore investigates trading strategies that are long only or short only (with the long or short positive being based on the same unconditional mean signal that was used in Panel A). The only difference being that if the signal is unfavourable, then the investor goes long in the risk-free rate for that day. The long only strategies perform similarly to the Table A strategies and are positive in only 19 and 9 (or 45% and 21%) strategies for the aggregate approach and coin-wise approaches, respectively.

The performance of the short only strategies however, as reported in Panel B of Table 4, are demonstrably different and are where the strongest and most persistent time-series association between return and the moments lies. Specifically, the aggregate short-RF strategy is positive in 34 (81%) of the strategies, with the highest returns being for the machine learning weighting (ML- $m_{1,2,3,4}$) of 11.7% p.a., followed by the GW- $m_{1,3}$ with 10.3% p.a. and the EW- $m_{1,2,3}$ of 8.3% p.a. Notably two effects seem to dominate, being the role of the 3rd moment (m_3), which features in each of the top seven trading strategies, including the pure moment weighting (which is 100% in m_3) which produces a return on its own of 7.0% p.a.. There is also a very strong mean-reversion component in these results, with m_1 featuring in eight of the top ten performing strategies and with the pure m_1 strategy producing a return of 6.3%.

For the Coin-wise short-RF strategy, the returns are noticeably lower on average, but again the machine learning weighting (ML- $m_{1,2,3,4}$) produces the highest return being 6.0% p.a., followed by the equal-weighted (EW- $m_{1,3}$) with 5.9% p.a. and the GW- $m_{3,4}$ with 4.7% p.a. As with the aggregate results, the 3rd moment plays a significant role, appearing in each of the top five trading strategies. Interestingly dispersion is a much more important signal at the coin level, with the pure moment m_2 portfolio producing a return of 4.0% p.a. and the m_4 moment playing a role in three of the top four strategies. Notably, at the coin level, the returns from a mean reversion strategy (short on the pure m_1 portfolio) is almost zero (a mere 0.46%), indicating that this mean-reversion affect is actually enhanced from the formation of portfolios.

Table 4 – Panel B: Conditional long-short trading strategies using lagged moment combinations

	(1) Utility Metric (U)	(2) Aggregate Long or RF (p.a.)	(3) Aggregate Short or RF (p.a.)	(4) Coin-wise Long or RF (p.a.)	(5) Coin-wise Short or RF (p.a.)
<i>Panel A: PM</i>	m ₁	0.036*	0.063**	-0.0209	0.0046
	m ₂	-0.016	0.0366*	-0.0157	0.0397*
	m ₃	0.0282*	0.0704**	0.0019	0.045**
	m ₄	0.0071	0.0557**	-0.0346	0.0151
<i>Panel B: EW</i>	m _{1,2}	-0.0234	0.018*	-0.0073	0.0345*
	m _{1,3}	0.0453**	0.079**	0.0248*	0.059**
	m _{1,4}	0.0358*	0.0702**	-0.0076	0.0258*
	m _{2,3}	-0.023	0.0266*	-0.0389	0.0108
	m _{2,4}	-0.0038	0.0488**	-0.031	0.0233*
	m _{3,4}	0.0309*	0.0744**	-0.0304	0.0121
	m _{1,2,3}	0.0396*	0.0835**	-0.0078	0.0346*
	m _{1,2,4}	0.005	0.0506**	-0.0216	0.0253*
	m _{1,3,4}	0.0395*	0.0774**	0.0108	0.0457**
	m _{2,3,4}	0.0079	0.0567**	-0.0207	0.0289*
	m _{1,2,3,4}	0.0009	0.0451**	-0.0382	0.0053
	m _{1,2}	-0.0168	0.0111	0.0101	0.0371*
<i>Panel C: GW</i>	m _{1,3}	0.0721**	0.1031**	0.0074	0.0355*
	m _{1,4}	0.0338*	0.0618**	-0.0209	0.0046
	m _{2,3}	0.0044	0.0302*	-0.0212	0.0039
	m _{2,4}	-0.0511	-0.0248	-0.0318	-0.0078
	m _{3,4}	-0.0114	0.0344*	0.0028	0.047**
	m _{1,2,3}	-0.013	0.0147	0.0021	0.0294*
	m _{1,2,4}	-0.0448	-0.0163	0.0043	0.031*
	m _{1,3,4}	0.0218*	0.0486**	-0.0047	0.0226*
	m _{2,3,4}	0.0049	0.031*	-0.0211	0.0031
	m _{1,2,3,4}	-0.0131	0.015	-0.0053	0.0218*
	m _{1,2}	-0.004	0.024*	-0.075	-0.0391
	m _{1,3}	-0.0337	-0.0064	-0.1393	-0.0542
<i>Panel D: FF</i>	m _{1,4}	-0.0165	0.0103	-0.131	-0.0272
	m _{2,3}	-0.0207	0.0086	-0.0716	-0.0526
	m _{2,4}	-0.0158	0.0333*	-0.0949	0.0246*
	m _{3,4}	-0.0358	-0.0034	-0.1435	-0.0787
	m _{1,2,3}	-0.0003	0.033*	-0.0856	-0.041
	m _{1,2,4}	-0.0534	-0.0185	-0.1092	-0.0339
	m _{1,3,4}	-0.0126	0.0209*	-0.1013	-0.0235
	m _{2,3,4}	-0.0496	-0.0096	-0.0637	-0.0093
	m _{1,2,3,4}	-0.0731	-0.0461	-0.1257	-0.0437
	m _{1,2,3,4}	0.081**	0.1172**	0.0255*	0.0603**
	m _{1,2,3,4}				
	m _{1,2,3,4}				
<i>Panel E: ML</i>	m _{1,2,3,4}				

Note: This table presents contemporaneous correlations of the first L-moment (return) with portfolios of moment combinations for each of four different weighting methods. Note that the daily L-moments for each cryptocurrency is calculated from intraday data and is equally weighted by coin. Panel A reports the Pure moment (PM) portfolios, whereby the weight in each moment is 100%, while Panel B weights each moment equally (EW), as per equation (5). Panel C uses the GUN weights (GW) of Billio et al (2022) (as per equation (6)), Panel D uses the Fama and French (FF) double and triple sorting procedures to form the moment portfolios while Panel E uses the machine learning (ML) approach to determine the weights on each moment. The symbol * indicates that the return for the strategy is in excess of the risk-free rate, while ** indicates that the return is in excess of the buy and hold position on the equal-weighted cryptocurrency portfolio.

5. DISCUSSION AND CONCLUSION

The primary objective of the paper was to document and analyse the time series behaviour present in the distribution of cryptocurrency returns. In doing so, we sought to select a methodology which was consistent with theory, for which we adopted the generalised utility of moments, which provided us with a relatively bias-free framework from which to evaluate this proposition. The initial results demonstrated a lack of evidence for the traditional finance risk-return relation (between the first two moments), but a significant role for asymmetry, as represented by the 3rd moment, having a highly significant relation to the mean when measured contemporaneously.

The trading strategy results were particularly interesting, and our conclusions are as follows. First, in relation to the choice of weighting for the moments, while the machine learning and GUN weightings are superior, the choice of weighting structure is not driving the results and indeed equal-weighted portfolios performed consistently well overall. Second, the aggregate results were universally higher than the coin-wise strategies, indicating that these patterns are systematic, rather than diversifiable and as with traditional financial instruments, cryptocurrencies investors who adopt a portfolio approach can obtain returns well above the risk-free and buy and hold benchmarks. Third, while the long-short strategies performed modestly, the key take-away was the performance of the short-side strategies, which produced a positive return in over 80% of strategies, regardless of the moment combination or weighting method used. This finding adds to the growing literature (such as Lee and Wang (2024)) in relation to signed tests of predictability in the cryptocurrency context. Fourth, consistent with the contemporaneous results discussed earlier and with the theoretical propositions of Kraus and Litzenberger (1976) and Harvey and Siddique (2000), the role of asymmetry is significant at the aggregate level. While the return can be enhanced by combining with other moments, strategies based solely on the 3rd moment performed very well. Finally, the returns for cryptocurrencies display significant negative serial correlation at the daily return frequency. In a trading strategy the implication of this is that a contrarian (mean-reversion based) strategy is more appropriate in these markets, rather than the momentum strategy that it typically adopted for traditional securities at this frequency.

These findings raise a number of questions for future research. First, machine learning was able to identify a more economically significant set of moment weights than the theoretical and naïve models. While beyond the scope of the present paper, this represents a fruitful area of research. Secondly, our allocation approaches were all based on static weights, and this significantly limited the performance of the trading strategies. Were the focus of the paper on identifying the most profitable trading strategy, a dynamic weighting framework would no doubt produce much higher returns. Indeed, if we were to assume perfect estimation precision with such an approach, the returns from the buy and hold strategy across our sample would be just under 300% per annum. Finally, while we adopted the theoretical framework of Hong and Stein (1999) to underpin our trading strategy, we made no attempt to differentiate whether fluctuations in moments were due to information events or patterns in trading behaviour. This distinction and subsequently delineating the trading strategy along these lines would not only be intuitively appealing and informative but would likely be better able to exploit the mean-reversion characteristics that we observe in cryptocurrency returns.

6. REFERENCE LIST

- Almeida, J. and T. C. Gonçalves (2022). "A Systematic Literature Review of Volatility and Risk Management on Cryptocurrency Investment: A Methodological Point of View." Risks **10**(5).
- Baek, C. and M. Elbeck (2015). "Bitcoins as an investment or speculative vehicle? A first look." Applied Economics Letters **22**(1): 30-34.
- Bernard, P., et al. (2019). "A financial fraud detection indicator for investors: an IDeA." Annals of Operations Research.
- Billio, M., et al. (2022). "A meta-measure of performance related to both investors and investments characteristics." Annals of Operations Research **313**(2): 1405-1447.
- Campbell, J. Y. (2000). "Asset Pricing at the Millennium." The Journal of finance **55**(4): 1515-1567.
- Chen, R., et al. (2022). "Can salience theory explain investor behaviour? Real-world evidence from the cryptocurrency market." International Review of Financial Analysis **84**.
- De Bondt, W. F. and R. Thaler (1985). "Does the stock market overreact?" The Journal of finance **40**(3): 793-805.
- Fama, E. F. and K. R. French (1992). "The cross-section of expected stock returns." The Journal of finance **47**(2): 427-465.
- Gordon, M. J. (1959). "Dividends, earnings, and stock prices." The review of economics and statistics: 99-105.
- Harvey, C. R. and A. Siddique (2000). "Conditional skewness in asset pricing tests." The Journal of finance **55**(3): 1263-1295.
- Hayes, A. S. (2017). "Cryptocurrency value formation: An empirical study leading to a cost of production model for valuing bitcoin." Telematics and Informatics **34**(7): 1308-1321.
- Hong, H. and J. C. Stein (1999). "A Unified Theory of Underreaction, Momentum Trading, and Overreaction in Asset Markets." The Journal of finance **54**(6): 2143-2184.
- Jacobs, H. (2015). "What explains the dynamics of 100 anomalies?" Journal of Banking & Finance **57**: 65-85.
- Jegadeesh, N. and S. Titman (1993). "Returns to buying winners and selling losers: Implications for stock market efficiency." The Journal of finance **48**(1): 65-91.

Jia, Y., et al. (2021). "Higher moments, extreme returns, and cross-section of cryptocurrency returns." Finance Research Letters **39**.

Karagiorgis, A., et al. (2023). "The Skewness-Kurtosis plane for cryptocurrencies' universe." International Journal of Finance & Economics.

Khuntia, S. and J. K. Pattanayak (2018). "Adaptive market hypothesis and evolving predictability of bitcoin." Economics Letters **167**: 26-28.

Kraus, A. and R. H. Litzenberger (1976). "Skewness preference and the valuation of risk assets." The Journal of finance **31**(4): 1085-1100.

Kyriazis, N., et al. (2020). "A systematic review of the bubble dynamics of cryptocurrency prices." Research in International Business and Finance **54**.

Lee, S. S. and M. Wang (2024). "Variance Decomposition and Cryptocurrency Return Prediction." Journal of Financial and Quantitative Analysis: 1-60.

Liu, Y., et al. (2021). "Risks and Returns of Cryptocurrency." The Review of Financial Studies **34**(6): 2689-2727.

MacGillivray, H. and K. Balanda (1988). "The relationships between skewness and kurtosis." Australian Journal of Statistics **30**(3): 319-337.

Pinto, J. E., et al. (2018). "Equity valuation: A survey of professional practice." Review of Financial Economics **37**(2): 219-233.

Rognone, L., et al. (2020). "News sentiment in the cryptocurrency market: An empirical comparison with Forex." International Review of Financial Analysis **69**.

Sapuric, S., et al. (2020). "The relationship between Bitcoin returns, volatility and volume: asymmetric GARCH modeling." Journal of Enterprise Information Management **35**(6): 1506-1521.

Smales, L. A. (2022). "Investor attention in cryptocurrency markets." International Review of Financial Analysis **79**.

Tay, A., et al. (2006). "Direction-of-Change Forecasts Based on Conditional Variance, Skewness and Kurtosis Dynamics: International Evidence No."

Xi, D., et al. (2020). "Investigating the Investment Behaviors in Cryptocurrency." The Journal of Alternative Investments **23**(2): 141-160.

8. APPENDICES

Table A-1: Description of cryptocurrencies and sample details

(1) <i>Symbol</i>	(2) <i>Name</i>	(3) <i>Market</i>	(4) <i>Trading volume</i>	(3) <i>Sample start</i>	(4) <i>Daily</i>
ADA	Cardano	289670863	5395451	18/04/2018	1164
ALGO	Algorand	7432793	373451	22/06/2019	742
ATOM	Cosmos	302999220	8355012	29/04/2019	794
AVAX	Avalanche	70718512	12107707	22/09/2020	294
BCH	Bitcoin cash	15050473630	1110770120	28/11/2019	587
BNB	Binance	110619000000	2661779631	7/11/2017	1320
BTC	Bitcoin	53353500000000	1165860000000	18/08/2017	1399
BTG	Bitcoin gold	17582642	2443180	16/04/2021	94
CELO	Celo	31501194	246543	5/01/2021	193
DASH	Dash	9214578250	93416434	28/03/2019	826
DCR	Decred	84715965	2295238	30/07/2020	348
DGB	DigiByte	85381	10185	20/07/2020	358
DOGE	Dogecoin	76576141	2669463	5/07/2019	729
DOT	Polkadot	2095180974	74209386	19/08/2020	328
EGLD	Electronic Gold	797081469	41236777	3/09/2020	313
EOS	Electro-Optical	208591675	16067066	29/05/2018	1123
ETC	Etherium	3466091475	119647799	12/06/2018	1109
ETH	Etherium	2537950000000	33409914080	18/08/2017	1399
FIL	Filecoin	2923149534	123806040	16/10/2020	271
HBAR	Hedera	1419979	65060	29/09/2019	644
HNT	Helium	35571557	806590	24/09/2020	292
ICP	Internet	1571862068	39483021	11/05/2021	71
KLAY	Klaytn	1699830	2978	24/06/2021	27
KSM	Kusama	3434411832	114915850	4/09/2020	312
LTC	Litecoin	7345111705	505664451	14/12/2017	1283
LUNA	Luna	218808741	6358635	21/08/2020	326
MATIC	Polygon	268935361	3004694	27/04/2019	796
MDX	Madala	13344683	44486	24/05/2021	58
NEAR	NEAR Protocol	12566579	775304	14/10/2020	273
NEO	Neo	461920991	55052868	21/11/2017	1306
ONE	Harmony	563570	47489	1/06/2019	762
ONT	Ontology	7205137	1064468	8/06/2018	1113
QTUM	Qtum	31043956	5568923	20/03/2018	1193
RVN	Ravencoin	330819	38018	25/09/2019	648
SC	Siacoin	67934	6416	6/07/2020	372
SOL	Solana	2343280057	28408462	11/08/2020	336
STX	Stacks	3431320	101748	25/10/2019	619
TFUEL	Theta Fuel	12281371	131724	24/05/2019	770
THETA	Theta	324078932	7214964	10/04/2019	813
TRX	Tronix	1964418	132611	12/06/2018	1109
VET	VechainThor	4469698	198863	25/07/2018	1068
WAVES	Waves	89164566	4703630	18/01/2019	894
XEM	New Economy	2815329	123352	24/11/2020	232
XLM	Stellar	5703959	478199	1/06/2018	1120
XMR	Monero	4147573053	78894374	15/03/2019	839
XRP	Ripple	101885823	6007224	5/05/2018	1147
XTZ	Tezos	20146583	1780677	24/09/2019	649
ZEC	Zcash	1965841140	78775494	21/03/2019	833
ZEN	Horizen	275905389	11595510	6/07/2020	372
ZIL	Zilliqa	429429	43715	19/02/2019	862

Note: This table presents descriptive statistics for each of the 50 cryptocurrencies in our sample. The values in the table represent daily figures, where each value is calculated from intra-day data for each currency. Columns (1) and (2) report the cryptocurrency symbol and name, (3) the market capitalisation (number of coins time price), taken as the value at the sample end date (21st July, 2021) and expressed in USD and (4) the average daily trading volume on the exchange (in USD) over the sample, (5) and (6) denote the first day that price data is available for that cryptocurrency and the corresponding number of daily observations in the sample.